MY 1st PROGRESS REPORT

Agent Capability Graph

Building an RDF Knowledge Graph to Discover MCP Servers and Agent Skills

Main Supervisor:	Raphael Troncy
Secondary Supervisor:	Thibault Ehrhart
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1 Abstract

The *Agent Capability Graph* project, which I am working on with my teammate, aims to study whether an RDF-based knowledge graph can improve the discovery of *Model Context Protocol* (MCP) servers and *Agent Skills*. Artificial intelligence agents can use external tools and specialized skills, but choosing the right resource becomes difficult when their number increases and their descriptions are heterogeneous.

With my teammate, the idea is to move from a simple list of servers and skills to a structured representation of their functions, parameters, resources, dependencies, access requirements, risks, and information sources. The graph could then be queried to answer capability-oriented questions, for example finding a tool capable of querying a SPARQL endpoint or identifying servers that access the file system.

This first phase of the work for me is primarily to understand the concepts of MCP, Agent Skills, RDF, and SPARQL, and to examine possible data sources. A first direction consists in building a limited, traceable, and evaluable prototype before considering broader coverage.

2 Problem Understanding

2.1 Why Is Capability Discovery Difficult?

MCP server registries and Skill repositories describe resources with varying formats and levels of detail. A keyword search may miss a tool if its description uses different terminology from the query. Tags depend on the categorization made by contributors. Embeddings can find semantically close descriptions, but they do not necessarily explicitly represent parameters, permissions, or possible effects of a tool.

The problem is therefore not only to find a name matching a query. It is about understanding *what a capability allows to do*, *under what conditions it can be used*, and *what precautions its usage requires*.

2.2 Project Objective

Our objective is to create an RDF knowledge graph prototype describing a selected corpus of MCP servers and Agent Skills. The graph must connect servers, tools, skills, parameters, resources, software projects, risks, quality signals, and data sources.

The project does not aim only to build a search engine. It seeks to evaluate whether an explicit semantic representation allows for more relevant, more explainable, or more precise discovery than simpler methods.

2.3 The Fundamental Question of Our Project

> **Can a semantic capability graph improve the discovery and selection of MCP servers and Agent Skills compared to traditional keyword search, tag-based search, or embedding-based search?**

For me, this question involves two dimensions:

- construction: collecting, normalizing, and representing metadata in a coherent RDF graph;

- evaluation: defining search tasks and comparing the graph results with baseline methods.

3 Technical Concepts of Our Project

3.1 MCP: Model Context Protocol

MCP is an open protocol that standardizes how an AI application connects to external tools and context sources. In the general architecture, a host contains the AI application, an MCP client manages communication, and an MCP server exposes features.

Servers can notably expose:

- **tools** that the model can request to execute;
- **resources** that provide data or context;
- **prompts** that offer instruction models or workflows.

3.2 Agent Skills

An *Agent Skill* is an organized set of instructions, resources, and possibly scripts that helps an agent accomplish a task or a family of tasks. A Skill can contain a `SKILL.md` file, auxiliary scripts, and reference documents. The agent can load these elements when the context requires it.

It is necessary to distinguish a Skill from an MCP server. A Skill mainly provides procedural knowledge and reusable instructions; an MCP server exposes features accessible through a protocol. Both can be complementary: a Skill can explain a procedure while an MCP tool carries out an external operation.

3.3 RDF: Knowledge Representation

RDF (*Resource Description Framework*) is a standard model that represents information in the form of triples:

```
subject   predicate   object
```

These triples are modeling examples, and not data already collected. Each triple expresses a relation; their accumulation forms a knowledge graph.

3.4 SPARQL: Querying the Graph

SPARQL is a query language for RDF data. It allows searching for resources by describing expected triple patterns. A query could, for example, search for tools associated with file manipulation. The exact query will depend on the chosen schema and the present data.

4 Proposal for a Pipeline

An initial breakdown of the work could be as follows:

1. **Select sources:** we will choose one MCP registry and 3 Skill repositories.
2. **Collect metadata:** retrieve just a few descriptions, names, versions, repositories, tool schemas, and `SKILL.md` files.

3. **Normalize:** harmonize fields, handle missing values, and identify duplicates.
4. **Model:** define the necessary RDF classes and relations.
5. **Build the graph:** convert normalized data into RDF and preserve their provenance.
6. **Validate:** apply SHACL constraints.
7. **Query:** write just a few small SPARQL queries answering discovery questions.
8. **Evaluate:** compare results with baseline methods.

5 Proposed Ideas for the First Phase

5.1 Limiting the Initial Scope

For me, it is better to start with a controlled corpus size rather than trying to collect a large amount of data. A first corpus could cover a few complementary domains: Web search, files, databases, and document processing. The final choice must take into account metadata availability, collection feasibility, quality, and capability diversity.

5.2 Separating Servers, Tools, and Skills

The model should distinguish at least:

- an MCP server, which exposes capabilities;
- an MCP tool, which corresponds to an invocable operation;
- an Agent Skill, which groups instructions and resources;
- a capability, which describes the satisfied functional need;
- a parameter, dependency, resource, or risk signal;
- a provenance source, which indicates where the information comes from.

This separation will avoid treating a server, a tool, and a Skill as interchangeable objects. Relations can indicate that a server exposes a tool or that a Skill helps perform a task.

5.3 Comparing to a Baseline Method

A first experiment could use the same corpus and the same set of queries to compare:

- keyword search;
- tag-based search;
- embedding-based search;
- structured search based on RDF and SPARQL.

The planned metrics are precision, recall, and top results quality. Relevant results will need to be defined for each query, followed by a qualitative error analysis.

6 Proposed Initial Work Plan

Here is a proposed plan:

Step	Activity	Expected Result
1	Targeted reading of MCP, Agent Skills, RDF, and SPARQL	Concept summary and glossary.
2	Inventory of registries and repositories	Comparative table of sources and their limits.
3	Selection of a small corpus and test queries	Documented corpus and representative questions.
4	Proposal of RDF model and constraints	Preliminary schema, example triples, and candidate SHACL constraints.
5	Prototype for collection, conversion, and querying	First RDF data and SPARQL queries.
6	Preparation of comparative evaluation	Protocol, relevance criteria, and metrics defined.

7 Current Progress and Next Actions

This report is an initial formalization of my initial understanding of the subject. The proposed next actions are:

1. read official introductions to MCP and Agent Skills and identify available metadata;
2. Understand more RDF because I don't have background on this field and learn SPARQL also;
3. inspect the MCP registry and a few Skill repositories;
4. Understand the part of collection and evaluation;
5. prepare a summary table of sources and discuss the scope with supervisors;
6. propose a first RDF schema and a few demonstration queries.

8 Conclusion

The project aims to improve capability discovery for AI agents by linking MCP servers and Agent Skills in a structured semantic representation. RDF could provide the representation model, SPARQL the search mechanism, SHACL the consistency checks, and existing vocabularies a basis for describing projects, categories, and provenance.

The priority for the first phase for us, as suggested by our supervisor during our first meeting, is not to immediately develop a complete system, but to define a relevant corpus, understand available metadata, design a model simple enough to be achievable, and prepare a credible evaluation. The value of the graph will then need to be demonstrated by comparison with simpler methods.